SENPAI: Sidereal Enriched Rate-Track Astrometry in Deep Imagery of Solar System Bodies*

J. Zachary Gazak,[1] Lauren Fisher,[2] Matthew Phelps,[1] Ryan Swindle,[1] Leonard Baruela,[2] and Justin Fletcher[1]

[1]*Space Systems Command A&AS*
[2]*KBR*

ABSTRACT

Rate-track imagery−in which a telescope tracks an object's proper motion instead of the stellar background−has long been used to enhance the signal-to-noise ratio of faint solar system bodies such as asteroids, comets, and artificial satellites. Images from such observations are characterized by point source targets and streaked stars. While astrometric plate solving−matching ubiquitous and unique star fields to a world coordinate system−is a mature field of research, the complexities of detecting and measuring the centroid of star streaks in rate track imagery decreases astrometric accuracy. This is unfortunate, given that defining and refining orbits of bodies of the solar system require accurate astrometric positions. In this work, we propose a novel observing methodology for experiments requiring the enhanced signal-to-noise ratio of rate track imagery and the astrometrics of orbiting solar system bodies by capping rate track sequences with a sidereal observation at the tracked object's position. The resulting data provide high-precision astrometry and a natural filter for detecting star streaks in the rate-track imagery. We demonstrate astrometric uncertainties in right ascension and declination of $\leq 1.38''$ and $\leq 0.96''$, respectively, on observations of calibration satellites using telescopes with pixel scales (instantaneous field's of view) of $0.90''$ - $1.68''$. This technique−**S**idereal **EN**riched **P**recision **A**strometric **I**ntelligence (SENPAI)−is designed to run automatically, requires no tuning or calibration frames on a per sensor basis, and throughput exceeds 85% on most sensors tested. SENPAI far exceeds the threshold requirement of $\leq 3.0''$ of uncertainty in object positioning for initial determination of−and state updates to−the orbits of artificial satellites.

Keywords: Detection (1911) — Artificial satellites (68) — Algorithms (1883) — Automated telescopes (121) — Surveys (1671)

1. INTRODUCTION

We introduce a concept of operations combining the photometric depth of rate-track imagery and the astrometric precision of sidereal imagery. This is accomplished by collecting a number of frames of rate track imagery utilizing a set of orbital elements (a state vector, for example, or a two line element set), and finishing the set of observations with a single exposure which begins on target but tracks the stellar background. This observing technique and algorithm, **S**idereal **EN**riched **P**recision **A**strometric **I**ntelligence (SENPAI), outputs star streak localization and astrometric plate solutions for each rate track frame. The high precision and throughput enables autonomous telescope operations used to detect and track artificial satellites and solar system bodies.

Further, on narrow field of view (FoV) telescopes, the ability to plate solve is limited in rate-track operations by the number of stars present in any given exposure. When tracking non-sidereal rates, the integrated counts from each star are spread over many pixels, effectively suppressing the limiting stellar magnitude relative to that of the tracked object(s) in frame. The SENPAI sidereal frame reverses this paradigm, collecting deep photometric data on the stellar background and allowing astrometric fits to narrow fields of view. In this frame, many additional stars are detected which remain below the noise floor in rate track exposures.

In this paper we demonstrate the technique by observing calibration satellites in Earth orbit on four small aperture (0.35−0.5 meter) telescopes with different cam-

Corresponding author: J. Zachary Gazak
jzg@alum.mit.edu

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Table 1. Telescopes used to benchmark SENPAI

Aper. meters	**Camera**	**Pixels** megapixel	**FoV** degrees	**iFoV** ″/pix
0.35	FLI 400	4.2	0.51 x 0.51	0.89
0.35	ATIK Apx60	3.8	0.55 x 0.83	1.25
0.35	Alta F47	0.3	0.24 x 0.24	1.68
0.50	QHY 600P	7.7	0.60 x 0.40	0.90
1.0	COSMOS-66	67.2	2.06 x 2.06	0.91

eras, fields of view, and observing locations. Calibration satellites are used because their orbital elements are known to extraordinary precision and thus the precision of SENPAI can be effectively measured. In practice, the technique naturally extends to any non-sidereal mission.

In Section §2 we review existing techniques to exploit rate-track observations. In §3 we describe the SENPAI concept of operations and algorithm in depth, and provide the outcome of calibration satellite experiments across four telescopes in §4. Section §5 formulates the applicability of SENPAI to different missions (tracking rates), and telescopes (fields of view, limiting magnitudes). We offer concluding remarks and future avenues of research in §6.

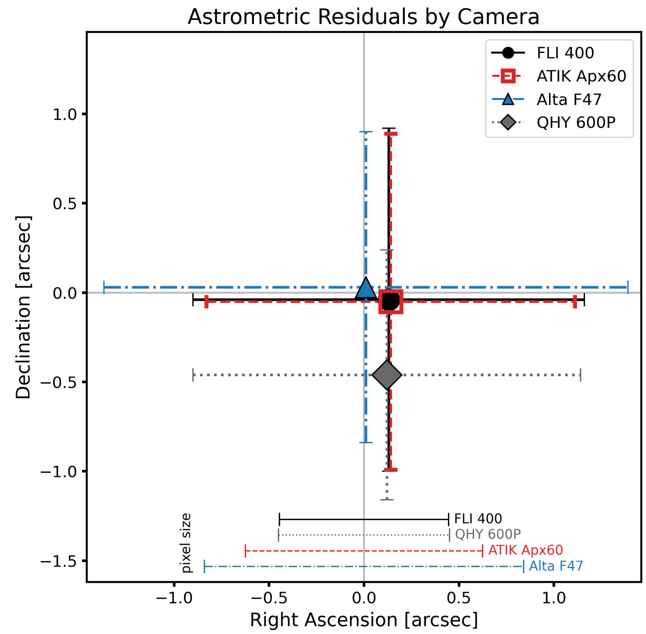

Figure 1. Astrometric precision of the SENPAI algorithm as measured against calibration satellite ephemeris for the four telescope / camera combinations studied in this work (Table 1, excluding the sparse data demonstration on the 1.0 meter system). Each camera's overall accuracy in right ascension and declination is plotted as a point, while error bars represent root mean square error (RMSE) of the measurements. Establishing and maintaining orbital parameters of earth orbiting satellites requires RMSE measurements less than 3″, a quality gate which SENPAI significantly surpasses on telescopes even in high seeing environments (Table 2).

2. PREVIOUS WORK

Non-sidereal tracking is a critical mode of operations for faint solar system bodies, in which signal-to-noise (SNR) is accumulated by tracking on the object's motion instead of the stellar background. With planned scientific missions such as the study of the first interstellar asteroid (Jewitt et al. 2017), careful manual star registration and stacking enables enhanced signal to noise via non-sidereal tracking. The age of astronomical survey missions−such as Pan-STARRS (Chambers 2018) and ATLAS (Tonry et al. 2018)−demand automatic data processing pipelines.

While such missions typically operate in sidereal mode and rely on stacked exposures to reach depths greater than the limiting magnitude of a single frame, the automated discovery of the faintest solar system bodies can be extended by either larger aperture telescopes−an expensive exercise−or by tracking on expected proper motions for given orbital parameters. For example, debris in Earth orbit can be studied more readily by tracking the expected orbital motion of that debris and exposing for longer periods of time on multiple small apertures than by consuming resources on larger mounts. This is the tactic employed by the space domain awareness (SDA) community (Sydney et al. 2000; Levesque 2011; Fletcher et al. 2019; Gazak et al. 2023). Similarly, non-sidereal deep stares of expected orbital parameters for solar system bodies such as asteroids, comets, and dwarf planets offer potential rich discoveries inaccessible by other methods, such as synthetic rate-track (Shao et al. 2014), which still suffer from noise accumulation and instrument inefficiencies such as readout time when short exposure stacking is employed.

Star streak detection in rate-track imagery is generally applied using only information present in a single frame and solved by signal processing algorithms carefully tuned to each sensor used (Levesque 2011). However, the techniques break down when few stars are present, the streak lengths are longer than tens of pixels, or the noise characteristics of a sensor change. Solutions to these issues, such as aligning and combining multiple frames to extend the star field as in Meredith et al. (2022), tend to introduce the same complications as streak detection in the preprocessing image tiling step, and still require streak detection. Recently, Felt & Fletcher (2024) proposed learned star detection in which neural networks are trained to detect and lo-

Table 2. SENPAI Performance and Observing Conditions

Camera	Dataset	Throughput	Seeing [″]	Residuals [″]		Residuals [pix]	
				RA	Dec	RA	Dec
FLI 400	72113	87.5%	5.03 ± 1.72	0.13 ± 1.03	-0.04 ± 0.96	0.15 ± 1.16	-0.04 ± 1.08
ATIK APX60	95176	88.3%	6.06 ± 1.74	0.14 ± 0.97	-0.05 ± 0.94	0.11 ± 0.78	-0.04 ± 0.75
Alta F47	95876	62.4%	4.99 ± 0.83	0.01 ± 1.38	0.03 ± 0.87	0.01 ± 0.82	0.02 ± 0.52
QHY 600P	79826	73.6%	3.66 ± 1.12	0.12 ± 1.02	-0.46 ± 0.70	0.13 ± 1.13	-0.21 ± 0.77

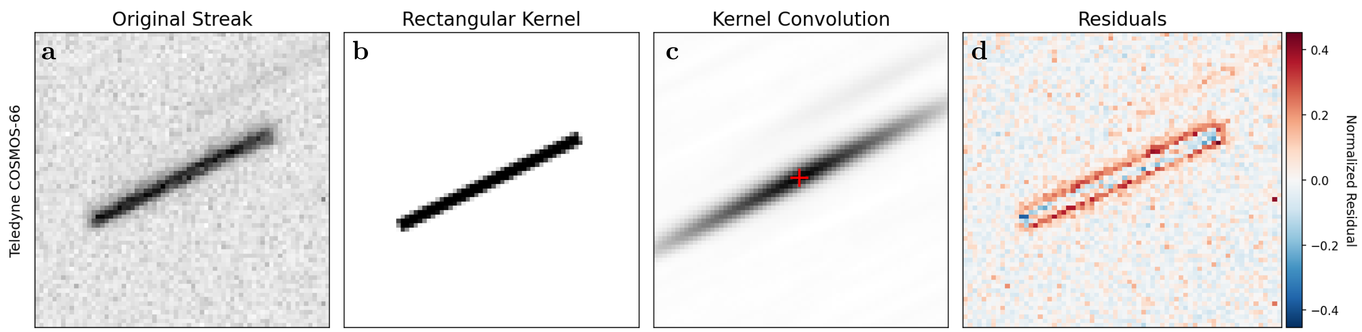

Figure 2. Demonstration of the convolution of a synthetic rectangular PSF and star streak. Note that the kernel does not model higher order effects (coma, along-streak turbulence, etc), but instead is designed to extract the geometric mean of a star streak rapidly and precisely.

calize streaked stars. These techniques offer the tantalizing benefit of rapid inference speed, which becomes critical for rapidly informing initial orbits of newly discovered targets, re-acquisition timelines, satellite maneuvers, etc. However, the burdens of compiling representative datasets for training such networks are considerable, requiring hundreds of thousands of frames that have been carefully human labeled. Further, data drift−both abrupt (new sensors) and gradual (aging sensors)−dictate the need for ongoing human labeling to tune learned algorithms.

To date, both classic and learned techniques to extract astrometric information from rate-track imagery require solutions that are implemented, tuned or trained, and maintained on a per-system basis. As such, these solutions do not scale with the number of telescopes employed in a network of survey telescopes. For large, collaborative networks of heterogeneous sensors (Fletcher & Kadan 2024; Cunio et al. 2022; Herz et al. 2017), entire software and data engineering teams are required to monitor and maintain the quality of these astrometric workflows.

3. METHODS

3.1. *SENPAI Observations*

In autonomous rate-track surveys, telescopes are tasked to take multiple rate-track frames (traditionally ∼6-20 to feed initial orbit determination models based on target orbital regime) while tracking the state vector of a target or survey area. The modifications required for SENPAI are simple: after the final rate-track frame is read out, the telescope switches to sidereal tracking mode and collects one additional frame at the same integration time.

The set of rate-track frames are characterized by a field of streaked star point spread functions (PSFs) of a single length (l_{star}) and orientation (θ_{star}), point source PSFs for any targets well aligned to the state vector, and streaked PSFs not matching l_{star} and θ_{star} for any non-sidereal sources not well represented by the state vector. The sidereal frame is photometrically deep on the star field, with more point source stars above the noise floor than in streaked rate-track stars. In this frame the state vector target is streaked at l_{star} and θ_{star} if it has sufficient signal to be observed. These modalities are visualized in Figure 3.

3.2. *Sidereal-Track Astrometry*

The sidereal frame contains rich information on the photometric response of the telescope and observing conditions, astrometric seeing, FoV, and−of course−astrometric localization. All of this information is easily extracted once an initial astrometric fit is made. In this work we employ an Astrometry.net implementation containerized for our analysis pipelines (Lang et al. 2010). We utilize the 5200-LITE star indices which pro-

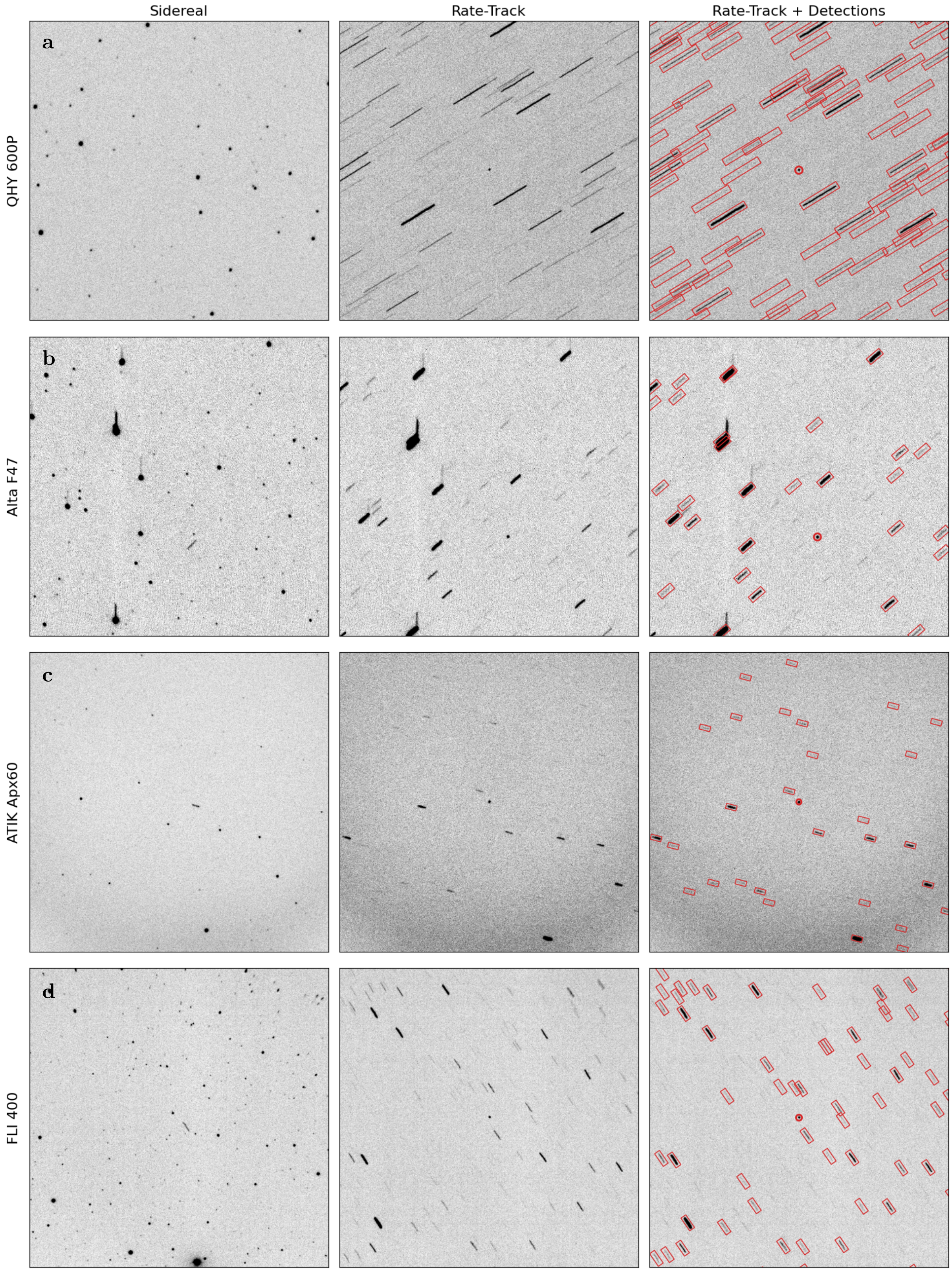

Figure 3. Example Sidereal, Rate-track, and SENPAI processed frames from each telescope used in this work. These frames are 512 x 512 pixel cutouts, which represents the full frame of the Alta F47 camera (row b), but only small regions of the remaining cameras (see Table 1). The first two columns represent the "rate-sidereal" mode of operations. The final column demonstrates the outputs from SENPAI: high astrometric precision star streak labels (red boxes) and object detections (red circles).

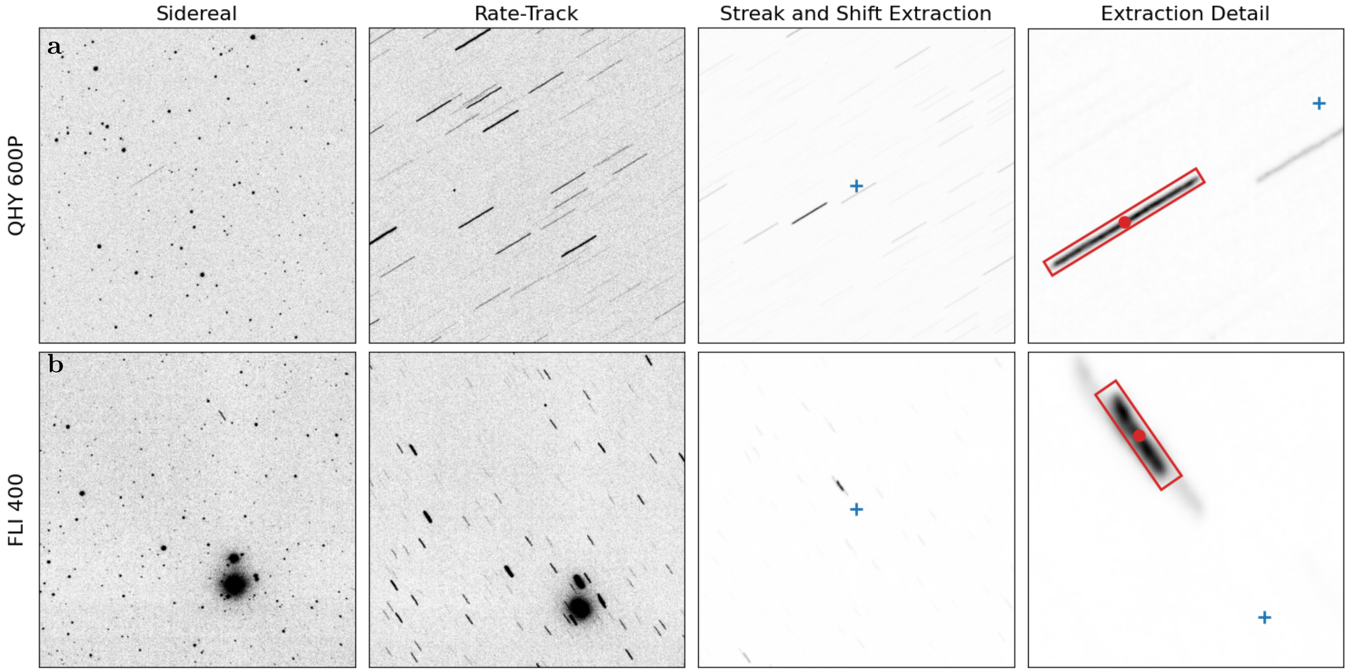

Figure 4. Extraction of the idealized streak point spread function, orientation, and shift from the sidereal frame. First column ("Sidereal"): sidereal frame with point source stars. Second column ("Rate-Track"): rate-track frame showing streaked stars. Third column ("Streak and Shift Extraction"): the cross correlation of the sidereal and rate-track frames, showing isolation of the streak length and orientation, with frame center (blue cross) annotated. Final column ("Extraction Detail"): Center detail of the third column, showing frame center (blue cross), and the ideal measured streak (red box) with center (red dot) easily extracted by the removal of all other meaningful signal from the frame. The shift between the stars in the sidereal and rate-track image is simply the vector between the red dot and blue cross.

vide speed and are performant for systems with up to 1 $\deg^2$ fields of view, and 4200 indices for larger formats. These indices combine the Tycho-2 and GAIA-DR2 catalogs (Høg et al. 2000; Gaia Collaboration et al. 2018).

The sidereal frame best encodes information on field distortion, PSF variability, and other instrumental effects including radial illumination falloff. While each of these effects are critical for photometry of stars and satellites in observed fields, only field distortion affects the SENPAI astrometric accuracy. Radial illumination falloff and PSF variability (apparent in Figure 8) for the systems presented here are caused by maximizing sky coverage and fully enclosing a telescope's imaging circle on a camera. The PSF variability presents as coma, increasing radially across the field and pushing flux outward, broadening point sources and streaks and producing characteristic skewness. However, because coma primarily redistributes light while leaving the location of the peak intensity nearly unchanged, the centroid of the streak is still well represented by the rectangular kernels introduced in this work.

We modify the streak kernel at each star location using the local sky-to-pixel Jacobian derived from the Astrometry.net sidereal world coordinate system's (WCS) simple imaging polynomial (SIP), which captures position-dependent variations in the projected streak direction and length. For radial distortions seen most commonly in this work, a 3rd order SIP provides precise corrections. This treatment accounts for field distortion and ensures that the kernel used in the matched filter is geometrically consistent with the local mapping from sky to detector. This set of corrections is applied only when significant field distortion (mainly pincushion, see Figure 9) is measured. We note that for the observatories used in this work (Table 1), even the 4 $\deg^2$ square degree system, these corrections amount to roughly 1 degree in rotation and a few percent modifications to streak length.

Sidereal frames without initial astrometric solutions (~30% of data collected for this work) are ignored by the remainder of the SENPAI pipeline. The bulk of these cases are indicative of unusable data: saturated frames collected too near sunrise or sunset, cloudy weather, or otherwise obscured conditions.

With successful sidereal astrometric solutions, SENPAI has access to instantaneous field of view (iFoV; the

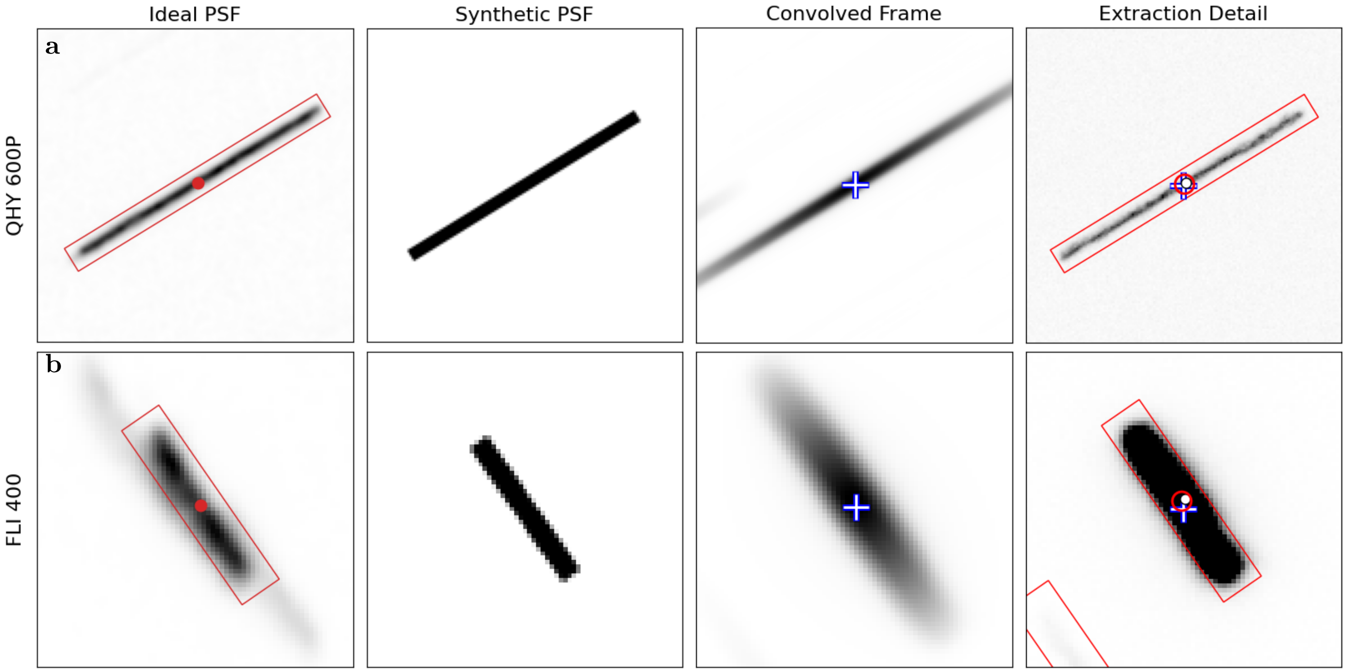

Figure 5. Detection of centers of streaked stars via convolution with a synthetic rectangular PSF. First column: The idealized streak from Figure 4, with center and extent (2 FWHM box) annotated. Second column: The synthetic PSF created from $(l, \theta)_{star}$ of the first column and the FWHM measured from the sidereal frame (see §3.2). Third column: The brightest point of $I_{rate,PSF}$, the rate track frame convolved with the synthetic streak PSF. The blue-white cross denotes the distinct maximum point of the pyramidal function formed by this convolution, and the streak center. Fourth column: The rate track image denoting the streak in the third column. The streak shows three measurements of center: the white dot denotes application of frame shift alone, the blue-white cross denotes the result of convolution based localization, and the red circle shows the location after a repeated astrometric fit of all blue-white crosses in frame (see §3.3 for discussion).

spatial extent of each pixel in the field), FoV, and a fit of right ascension and declination across the field. The sidereal astrometric fit allows for a query of catalog stars present in the frame. SENPAI uses catalog stars to fit two dimensional Gaussian profiles to bright, well isolated stars. The median full width at half maximum (FWHM) of these measurements provides a combined atmospheric and instrumental seeing.

With access to iFoV and seeing, SENPAI downscales each frame (sidereal and rate-track) such that the PSF FWHM falls across 3-4 pixels. In cases where reduction is needed, raw frames are median filtered over the necessary pixel scale and then downsampled. This provides a few positive effects: hot and dead pixels are largely eliminated, high seeing or oversampled FoVs are properly sampled for downstream tasks, and the pipeline executes faster on smaller images.

3.3. *Rate-Track Astrometry*

The initial phase involves transferring the sidereal WCS of §3.2 to the final rate-track image. At it's simplest, this involves cross correlating the sidereal (point source) frame with the rate-track (streaked frame), producing a correlation image, $I_{sidereal,rate}$, shown in Figure 4. This image contains two key features. First, the pixel shift $S_{x,y}$ which best aligns these two images becomes the dominant feature in $I_{sidereal,rate}$. Second, the PSF of this dominant feature is a clean, isolated representation of the rate-track streak orientation and length.

Extracting $(l, \theta)_{star}$ is a two phase process. First, using the $I_{sidereal,rate}$ frame, we normalize that frame so each pixel falls between 0 and 1, then use a flood-filling algorithm to select all pixels around the maximum pixel value until a border of pixels with values $\leq$0.5 is found. The points in the flood fill represent a rough proxy for FWHM. We extract an initial estimate of streak length l_{star} by simply taking the distance between flood fill endpoints, which are extracted as the minimum and maximum x and y points of the flood algorithm. For θ_{star}, we take the median angle formed between every flood fill point and the center of the frame.

While this initial measurement of $(l, \theta)_{star}$ is stable and generally successful, the frame normalization and flood fill to values $\leq$0.5 can be error prone in cases where the streak is not smooth or is saturated. Because of this, a second step is taken to use the streak length estimate to define a cutout box size, and then cutting out a number of the brightest streaks in the original rate track image. A median frame is calculated from the cutouts, which isolates and smooths the PSF. This cutout can then be more stably normalized, rotated to θ_{star}, and then compressed perpendicular to the streak direction. The resulting one dimensional boxcar function provides a stable and easily extracted measurement of streak length.

Once the shift ($S_{x,y}$) and PSF $(l, \theta)_{star}$ are extracted, a rotated rectangular 2D synthetic star PSF is generated and convolved with the rate-track image, producing a $I_{rate,PSF}$ image. Levesque (2011) point out that the convolution of two boxcar functions produces a pyramid with a distinct peak. Convolved rectangles produce 2D pyramids and, assuming the synthetic PSF is correct, the peaks of these provide precise streak centers in $I_{rate,PSF}$. False positive star detections are suppressed by limiting detections to 2×FHWM regions surrounding $S_{x,y}$ shifted positions from catalog stars in the sidereal astrometric fit. The sub-pixel coordinates of these detections can then be fed back into Astrometry.net to obtain a rate-track astrometric fit and WCS.

Characteristic residuals between the synthetic streak PSF and a star streak are illustrated in Figure 2. The rectangular streak PSF does not capture higher-order effects including coma and temporal turbulence; this is intentional. The kernel is designed to match the length and angle of star streaks, not provide a precise PSF fit, such that convolution is rapid across large fields. Although small coherent residuals appear along streak edges, the convolution produces a clean pyramidal maximum at the geometric center of the star streak.

There are three methods of transferring the high precision sidereal WCS to the rate-track frame. The fastest and lowest precision method is to **apply the $S_{x,y}$ directly**. In this method, the construction of a synthetic PSF, $I_{rate,PSF}$, extraction of rate star detections, and astrometric fitting are skipped. However, the pixel shift alone does not encode sub-pixel shifts, any changes in rotation of the field, and nonlinearity in the camera field. In addition, since a $S_{x,y}$ is always produced, even when a sidereal frame is clear and the rate track frame is obscured, this method can produce incorrect or illegitimate WCS and is not used. By requiring the detection of stars in the $I_{rate,PSF}$ the WCS is assured to be legitimate, and a higher quality WCS can be achieved by adjusting the sidereal WCS by **applying the sub-pixel rate-track star detections**. For the highest precision rate-track WCS, an additional 7-12 seconds is required to **perform an independent astrometric fit on rate-track detections**. In this work we report the slowest and highest precision SENPAI mode.

Once the synthetic PSF is determined, shifting the WCS to each remaining rate-track frame is trivial. First, two rate-track frames are cross-correlated, revealing a dominant and well isolated streak like feature, similar to that of the isolated streak in Figure 3, except instead of a star streak caused by cross correlating a sidereal and rate image, the feature in this image is the cross correlation of two streaks; a pyramidal PSF as in Figure 4. A visualization of this process is shown in Figure 5. The pyramidal feature provides a distinct peak, and the vector from frame center to that peak represents the shift in stellar centers from one rate frame to the next.

With a global frame shift measured, sub-pixel star streak locations are extracted by convolving the synthetic streak PSF of §3.3, and the WCS from one rate frame can be moved to the next using the global frame shift, bulk motions of each detected star streak, or by refitting a plate solution to the extracted streak centers.

3.4. *Frame Preprocessing*

The procedure for shifting a sidereal WCS to a rate frame (§3.2) and from rate to subsequent rate frame (§3.3) is enabled mainly by cross correlation operation on frames with similar features but global shifts. This operation is sensitive to three common image features: bright point source targets (satellites) in rate track frames, bright stars in only one frame in a pair (i.e. the bright star shifts in or out of frame), and cases where bright stars are cut off by a frame edge. In all of these cases, the dominant feature in the correlated image (last column of Figures 3, 4, and 6) is generally not the idealized offset between frames but a secondary feature governed by the rogue bright source. To reduce this failure mode considerably, each case is handled by preprocessing rate track frames.

For **extremely bright satellites**, the median of all rate track frames in a set is subtracted from each individual frame. This removes stable complex background features (fixed pattern noise or stray light), and reduces or removes bright, stationary sources (i.e. the satellite being tracked). In cases where features change rapidly (evolving thin cloud cover, or short period source variations due to, say, glinting or tumbling satellites) this step is imperfect, but still reduces the impact of bright point sources.

To remove **incomplete star streaks** cut off at frame's edge, a flood filling routine is used to isolate any areas of enhanced counts (2-σ_{counts} over the median subtracted image's mean counts) and filled with the image's

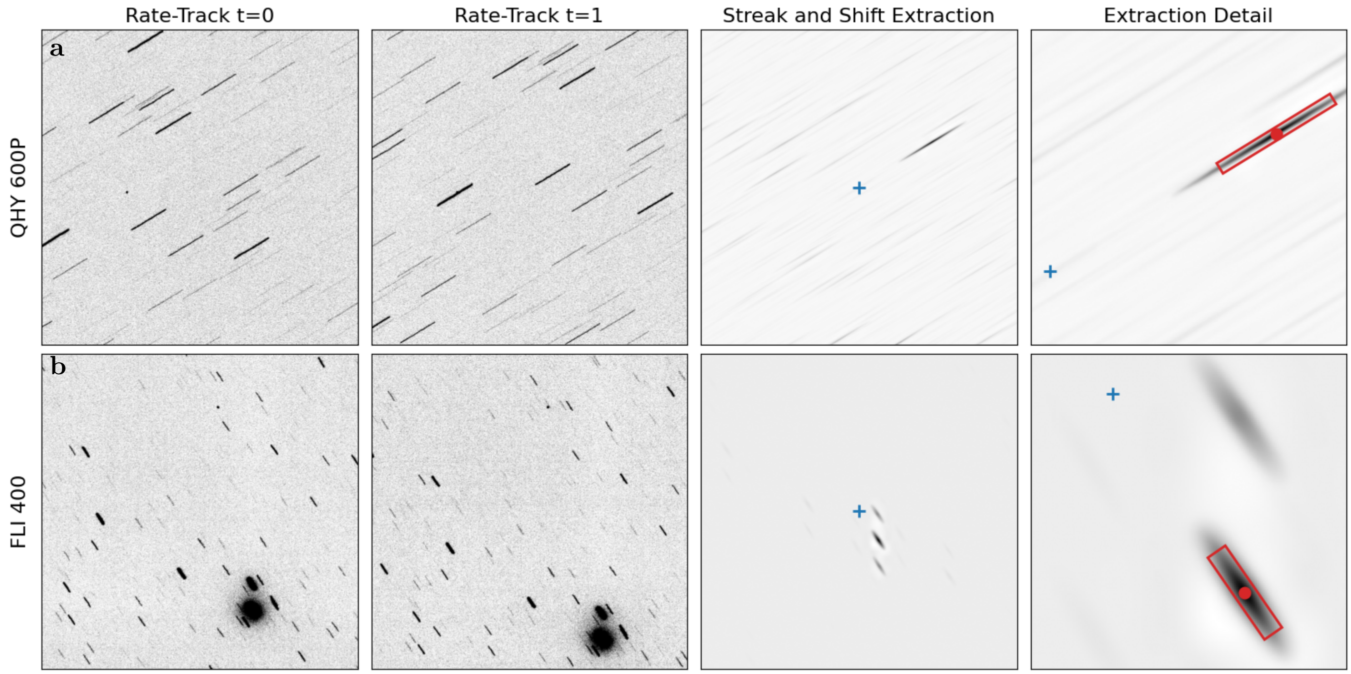

Figure 6. Shifting existing world coordinate system (WCS) between rate track frames. Unlike rate to sidereal shifts (Fig 3), both frame PSFs streaked. This provides a highly accurate streak orientation and shift (see columns 3 and 4, "Extraction"), as a pyramidal feature is created at the overall frame shift.

mean counts until the flood fill reaches a pixel boundary 1-σ_{counts} over the frame mean. This routine is applied until the outer border of the frame does not contain any bright pixels.

For **bright stars which leave the frame** between two subsequent frames, SENPAI enforces a percent difference in flux between frames and flood fills streaks starting with the brightest pixel in the counts-enhanced frame until such time as the two frames fall within a prescribed 10% difference. This preprocessing step does occasionally fail when a bright star leaves the frame and a similarly bright star enters the frame (but neither streak across the frame edge). In these cases, the percent difference between frames can be below the threshold, and the shift measured via cross correlating images can be the shift between these two bright stars, not the overall stellar background.

When these preprocessing modes fail and the measured shift is significantly incorrect, the SENPAI algorithm is then unable to detect star streaks around expected locations (§3.3), and the WCS refit algorithm fails. This failure mode decreases throughput but does not sacrifice astrometric quality.

4. RESULTS

4.1. *Dataset and Experimental Design*

SENPAI is tested on four cameras (CCD and CMOS sensors) and small telescopes with apertures between 0.35 to 0.5 meters. For demonstration purposes, we include figures and discussion from a much smaller dataset taken on a prototype larger aperture and FoV observatory (Swindle et al. 2024; Layden et al. 2025), but a full dataset is not yet available. The systems used are shown in Table 1. Each dataset contains between 70,000 and 100,000 frames collected at a rate of up to 2500 frames per night between June and August of 2024. Weather conditions are variable across the dataset, from photometric to obscured by cloud cover. Data from the 0.35m apertures was collected during catalog maintenance operations, and the 0.5m telescope was used to stress test SENPAI by varying exposure time from 0.5 to 10 seconds and collecting over an uncorrected range of star streak angles.

For the results of the SENPAI experiment (Table 2), no sensor- or telescope-specific settings are applied during the experiments. The algorithm is deigned to be largely sensor and system agnostic for fields of view between 0.1 and 5 degrees. Because SENPAI leverages the star field, issues can arise for sensors which are highly vignetted. This is easy to counter by applying a radial fit of expected vs actual signal to noise from stars in the sidereal exposure, and allowing SENPAI to mask the outer vignetted regions, however, this process is not applied by default.

4.2. *Throughput*

In Table 2, throughput shows the percentage of frames fit by SENPAI of those which are processable, and ideally could be fit. To accomplish this, we manually inspect each frame not fit by SENPAI, and flag them as processable if a human astronomer could reasonably measure a single star streak's length and orientation. Since our adopted astrometric plate solver from Lang et al. (2010) requires 8+ stars to complete a fit, SENPAI throughput is not expected to reach extremely high values. Still, this simple metric of "one clear star streak" is a significantly easier review process than requiring humans to count "8+ star streaks". In addition, future astrometric methods might require fewer stars, and modes of SENPAI which do not re-fit the plate solution for each rate track frame can operate on far fewer stars.

Under these conditions, SENPAI achieves 60-90% throughput on the datasets used. This number, which is highly dependent on individual dataset characteristics, is reported as a baseline metric for future experimentation.

4.3. *Astrometric Precision*

While throughput is important, astrometric precision is critical. To provide data destined for initial orbit determination, maneuver detection, and orbit updates for artificial satellites, the United States Space Force (USSF) requires that individual metric observations of RA, Dec, and time fall within a root mean square (RMS) error of 3.0″ regardless of sensor characteristics (United States Space Force, Space Delta 2 2023; Air Force Space Command 2019)

In this experiment, this result is measured using the roughly 10% of collects which are performed on calibration satellites whose positions are known with extraordinary precision. These ephemerides are accurate to between centimeters and meters, which is less than 0.006″ at geosynchronous orbit. With pixel iFoVs of roughly 1.0″, we ignore the miniscule error contribution from these ephemerides.

For each rate track frame processed by SENPAI, we measure a quadratic centroid of the point source closest to the expected satellite location (based on the SENPAI WCS and precise ephemeris. That extracted subpixel location is converted back into RA and Dec using the SENPAI WCS and compared to the known satellite location. Once the full set of measurements is complete, a global temporal offset is measured to calibrate the telescope clock, which typically has a millisecond level bias.

Using this technique, we measure astrometric precision of SENPAI generated WCS, and show RMS errors well below USSF's required 3.0″ for contributing sensors. We visualize bias and error measurements in Figure 1, and the values are tabulated in 2 for both pixel and arcsecond residuals. We demonstrate roughly 0.5 to 1 pixel RMS errors across all sensors, which translates to 0.7″ to 1.38″ RMS errors in Right Ascension and Declination measurements.

5. OPERATIONAL REGIME AND SYSTEM REQUIREMENTS

As currently designed, SENPAI relies on a "blind" astrometric fit on the sidereal frame, which requires ∼8 detected stars to return a ∼80% fit rate. While other techniques, including reliance on telescope pointing information and exploitation of spectral information (Phan et al. 2025) can reduce this requirement, a blind fit works on all imaging systems and is robust to errors in mount pointing models due to stale mount models, earthquakes, etc. To "transfer" a sidereal plate solution to a rate track frame, we require detection of three star streaks to enable measurement of field shift and rotation. Finally, streaks must be contained within an instrument's FoV to extract reliable centroids.

In this section we discuss the class of instruments that are applicable to the SENPAI technique as a function of sky coverage and desired tracking rates.

5.1. *Limiting Magnitude*

Streaking stars have reduced signal to noise ratios compared to sidereal tracked stars due largely to the extended background over which a streak falls. In this discussion, we consider the sidereal SNR,

$$\mathrm{SNR}_{\mathrm{sidereal}} = \frac{f\sqrt{T}}{\sqrt{bA_0}} \tag{1}$$

where f is the source flux (e^-/s or ADU/s) and b the background flux (e^-/s/arcsec^2 or ADU/s/arcsec^2) and A_0 represents the effective area of the PSF (here we assume a gaussian $A_0 = 2\pi\sigma^2$), and T represents exposure time in seconds. By assuming that a streaked star is simply the same sidereal PSF convolved along a line segment, we find that peak brightness of a streak is reduced by the gaussian error function,

$$I_{\mathrm{rate}}(L) = \frac{fT}{\sqrt{2\pi}\sigma L}\mathrm{erf}(\frac{L}{2\sqrt{2}\sigma}) \tag{2}$$

where I_0 represents the peak intensity and I_{peak} tends to I_0 when L $\ll \sigma$, and is inversely proportional to L as streak lengths far exceed PSF width,

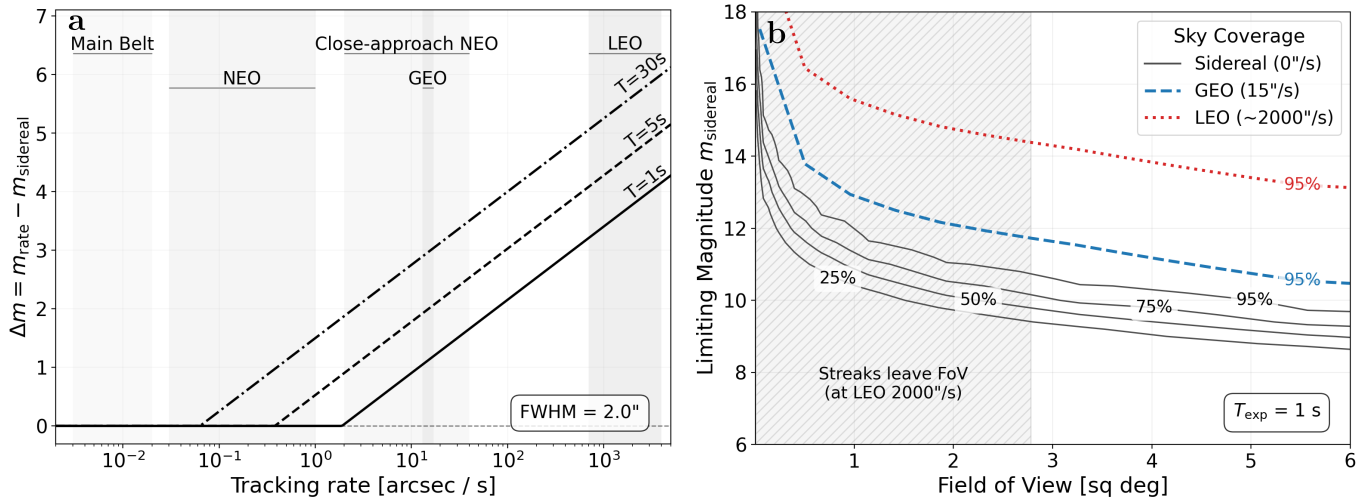

Figure 7. Operational Regime and System Requirements for SENPAI enabled telescopes. Panel a (left): the expected change in limiting magnitude for streaked stars in a field of view (FoV) as a function of tracking rate for a $2''$ FWHM system at 1, 5, and 10 second exposures. Panel b (right): Sky coverage contours as a function of sidereal limiting magnitude and FoV for a 1 second integration. For a given FoV, a telescope must have a limiting magnitude above (fainter than) the given line to reach that contour's sky coverage with SENPAI.

$$\frac{I_{\rm peak}(L)}{I_0} \Longrightarrow \simeq \begin{cases} 1, & L \ll \sigma, \\ \sqrt{2\pi}\,\sigma L^{-1}, & L \geq \sigma, \\ L^{-1} & L \gg \sigma. \end{cases} \tag{3}$$

We adopt a matched filter SNR which naturally arises from our rectangular PSFs and streak measurement, which requires modifying the SNR function by the tracking rate, r [$''$/sec],

$$\mathrm{SNR_{rate}} = \frac{\mathrm{f}}{\sqrt{\mathrm{brA_0}}} = \frac{1}{\sqrt{\mathrm{rT}}}\mathrm{SNR_{sidereal}} \tag{4}$$

provides a simple formula for the ratio of rate to sidereal SNR as a function of PSF width, tracking rate, and exposure time,

$$\frac{\mathrm{SNR_{rate}}}{\mathrm{SNR_{sidereal}}} \simeq 1.495\sqrt{\frac{\sigma}{\mathrm{rT}}} \tag{5}$$

and allows for a measurement of the affect of rate track operations on the limiting magnitude of a given system,

$$\Delta m = m_{\rm rate} - m_{\rm sidereal} = -2.5\log_{10}\left(1.495\sqrt{\frac{\sigma}{\mathrm{rT}}}\right) \tag{6}$$

which holds until rates grow small enough to no longer affect the shape of a sidereal PSF over the exposure time (in which case $\Delta m = 0$).

This function allows us to explore the affect on system limiting magnitude across various exposure times and tracking rates, and is visualized for a PSF of $2''$ in Figure 7a. Here we assume that the sidereal and rate track frames are exposed for the same amount of time, and see two effects. First, we notice that the Δm is larger for longer exposures – a natural effect of a streak falling over more background pixels while a sidereal PSF accumulates signal. Second, the Δm becomes significant for rapid rates, and systems are at risk of detecting too few rate track stars for astrometry.

5.2. *Sky Coverage*

We use a catalog combining Gaia, Tycho, and Hipparcos which is largely complete down to $\mathrm{m}_v = 18.0$ to calculate the limiting magnitude required for an observing system to reach the required number of stars within it's FoV to allow for blind sidereal and transferred rate astrometry. We do this by placing stars at given magnitudes in pixel locations on a large synthetic FoV, and convolving that field by a rectangle kernel of unity and of the shape of a given FoV. At each step, the outcome then is simply the number of stars in the field brighter than a given magnitude. This provides a smooth sky coverage function without the need for tiling or potentially missed windows.

We complete this experiment for FoVs between 0.01 and 6 deg^2, which well covers the range of telescopes studied in this work (which range from 0.26 to 4 deg^2).

The resulting map of sky coverage is then combined with the analysis of streak-reduced limiting magnitude to produce Figure 7b. In order to show the region where Low Earth Orbiting (LEO) satellites become inaccessible (as star streaks cover more than 1/3 of the frame

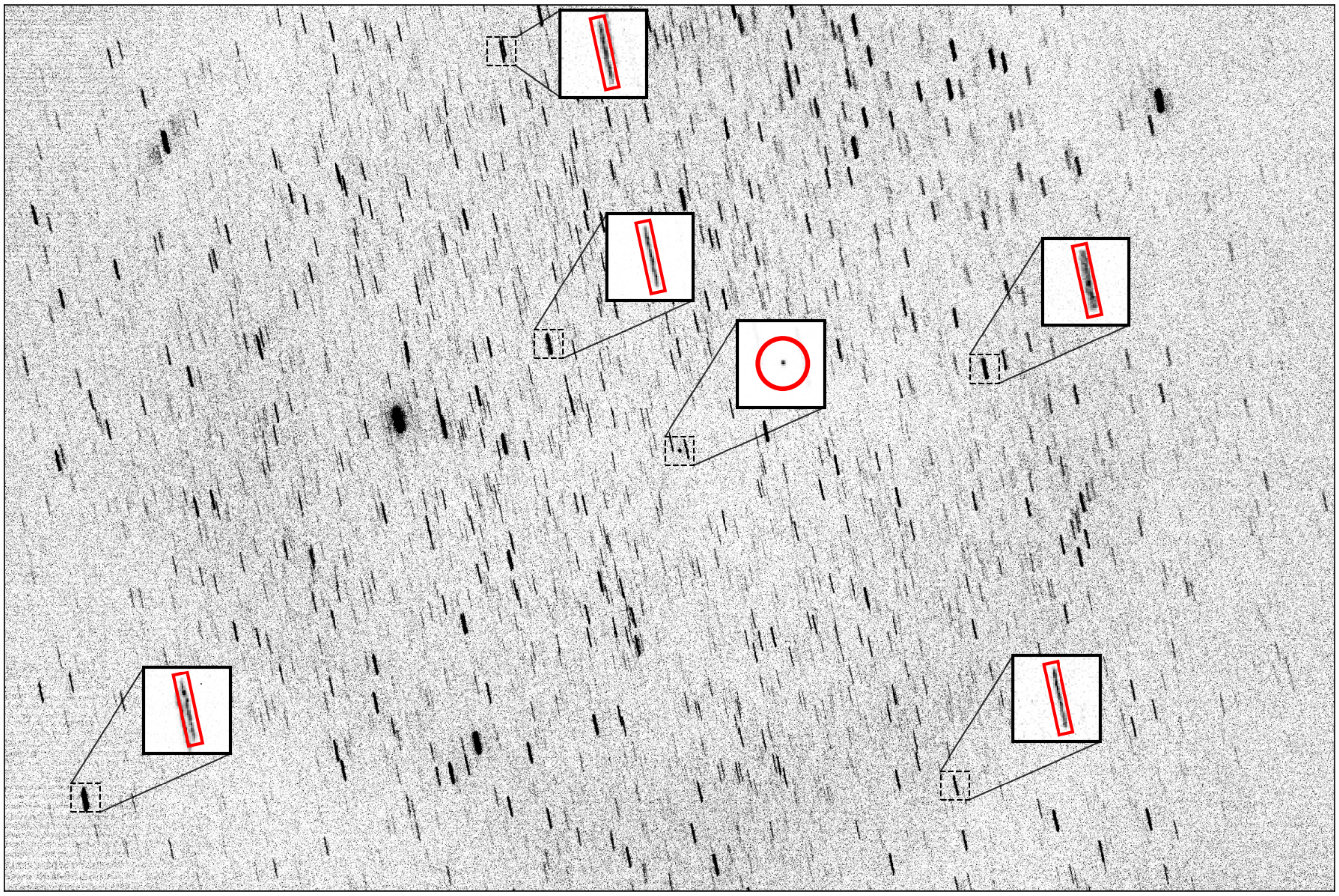

Figure 8. An example of the most crowded fields in the datasets studied in this work. This field is located at RA, Dec of 283.14 and -5.50 degrees, respectively, and lies along the galactic plane. SENPAI thrives in crowded fields as the correlation between sidereal and rate track frames has significant power around the frame shift and streak shape. This field also demonstrates signal falloff outside of the telescope's imaging circle, and the coma that occurs due to such an instrument design. SENPAI is largely unaffected by these types of variable PSF.

size) we produce this plot assuming exposure time T = 1 second. Longer than this and LEOs moving at ∼1-2 degrees per second require even larger FoV systems.

We first plot contours of SENPAI sky coverage for blind sidereal astrometry at 25, 50, 75, and 95% coverage as a function of FoV and sidereal limiting magnitude (again, in 1 second of exposure). On this map we add contours for 95% sky coverage for LEO and Geosynchronous (GEO) satellite tracking rates, which at 1 second require ~ 1 and 5 Δm performance to reach the 3 streaked stars required for SENPAI astrometry. As an example, consider the 1 meter, 4 deg^2 FoV system presented in this work (Table 1, Figure 9), to derive astrometry for LEO tracking against 95% of the night sky, the system must achieve a sidereal limiting magnitude of 14 in 1 second. For the same coverage at GEO track rates, the system would need only to achieve a limiting magnitude of 12. In practice, the 1 meter system regularly achieves 1 second limiting magnitudes far fainter, exceeding m=16 even in marginal weather.

6. CONCLUSIONS

We present a new concept of operations (rate-sidereal) and analysis suite (SENPAI) to enhance rate track operations for detecting and tracking space objects undergoing non-sidereal motion. By appending a sidereal exposure onto a set of rate track exposures, we demonstrate effective blind astrometric solving and high precision star streak measurements without the need for sensor specific calibration frames. These technologies enable expansive use of collaborative sensor networks without the need for dedicated processing chains per sensor. With SENPAI, sensors across industry, academia, and the military can be tasked against space domain awareness requirements when otherwise idle, and raw data consumed and converted into the observational information needed to build and update orbits for objects which move at non-sidereal rates, including primarily

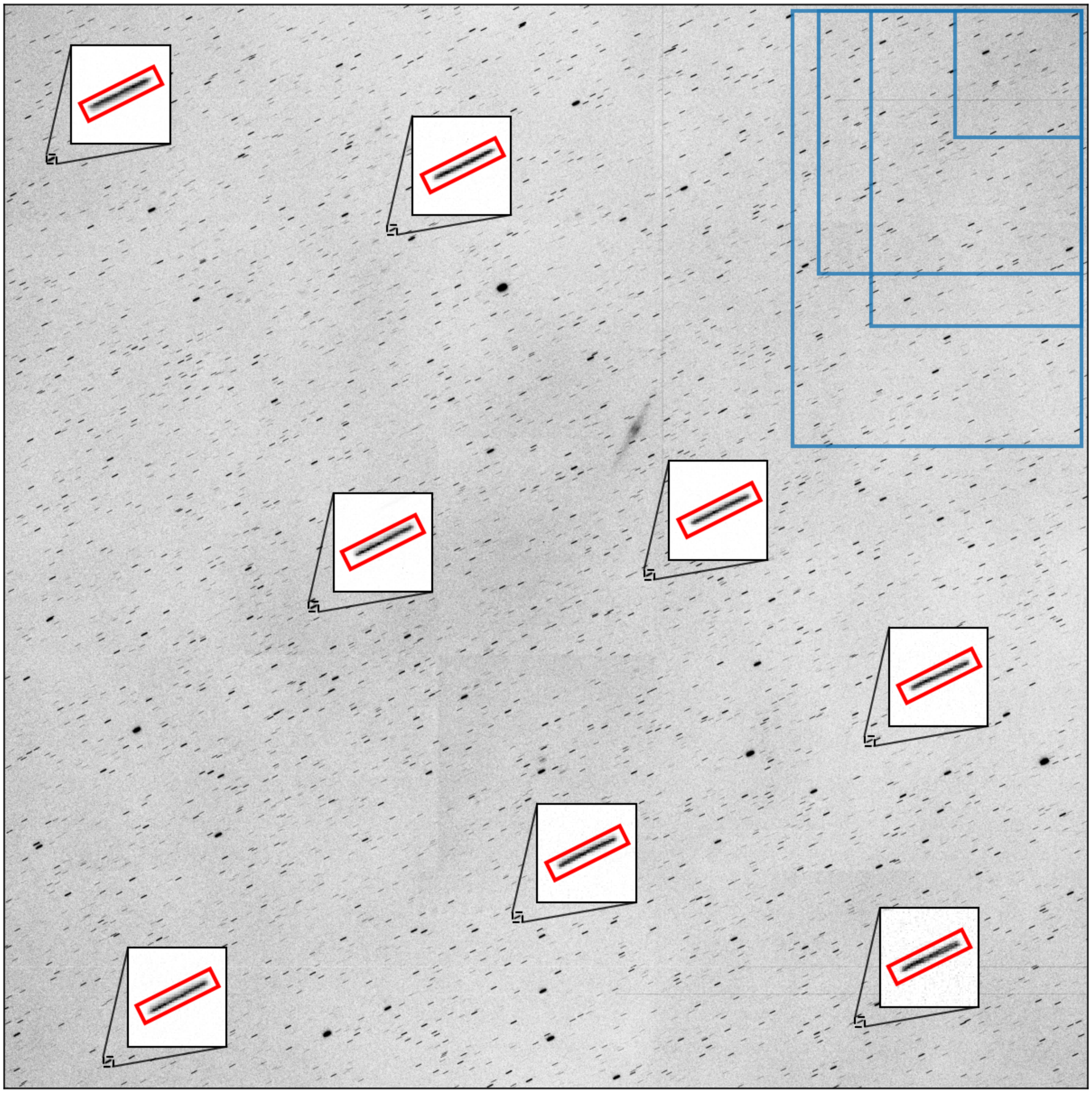

Figure 9. The 1 meter, 4 deg^2 field of view system proposed in Swindle et al. (2024) is capable of full sky GEO and LEO coverage using SENPAI. This example frame shows a number of enhanced streak images with SENPAI fits, as well as example overlays in blue of the rest of the telescopes studied in this work. This sensor is affected by pincushion radial distortions reaching $\sim$50 pixels in the corners. This effect is easily fit by a 3rd order SIP fit.

artificial satellites but with additional applications to asteroid discovery and tracking.

7. ACKNOWLEDGMENTS

This paper is approved for public release. Distribution is unlimited.

Software: Astrometry.net (Lang et al. 2010), Astropy (Astropy Collaboration et al. 2022, 2018, 2013), NumPy (Harris et al. 2020), Photutils (Bradley et al. 2024), PyEphem (Rhodes 2011), scikit-image (van der Walt et al. 2014) SciPy (Virtanen et al. 2020), SkyField (Rhodes 2019)

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